
Does the World Agree? A Model-vs-World Benchmark for Counterfactual Explanations of Time Series

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Abstract

Temporal counterfactual explanations are usually evaluated by whether they flip a classifier and remain close, sparse, or plausible relative to the observed input. These criteria do not establish whether the data-generating process would realise the proposed trajectory or its claimed outcome. We introduce a model-world benchmark built on additive-noise temporal structural causal models, where unit-level abduction is exact. The benchmark separately measures mechanism consistency (CF-faith), the number of timesteps that must be treated as interventions (do-complexity D), and the disagreement between the proposed and structurally replayed outcomes through an additive $A/B/C$ decomposition. We apply it to four temporal counterfactual baselines, a causal-residual foil, and two structural controls on three synthetic mechanism families. Conventional metrics frequently give a reassuring account, yet hard rollout CF-faith is zero in every baseline-family comparison, Pearl CF-faith is nonzero only for subsets of M-CELS proposals, and the claimed target-class rate exceeds the structurally realised rate in every baseline comparison. The observed disagreement is predominantly trajectory-level, and three baselines require near-complete intervention schedules. These results demonstrate a measurement gap in the reported synthetic benchmark realizations; they do not establish stable method rankings or transfer to unknown real-world mechanisms.

1 Introduction

A counterfactual explanation answers the question practitioners actually ask of a high-stakes sequential model: *what would have had to be different in this trajectory for the prediction to have been otherwise?* [Wachter et al., 2018]. In the temporal setting the question is routinely posed about clinical trajectories, industrial telemetry and financial series, and a substantial family of methods now answers it [Delaney et al., 2021, Ates et al., 2021, Höllig et al., 2022, Li et al., 2024, Paredes Cetina et al., 2025, Yan and Wang, 2023].

The evaluation protocol those methods are judged by has been remarkably stable: *validity* (does the counterfactual flip the classifier), *proximity* (how far is it from the factual), *sparsity* (how many entries changed), and *plausibility* (does it look like training data) [Delaney et al., 2021, Höllig et al., 2022, Pawelczyk et al., 2021]. Every one of those four quantities is a relation between the counterfactual and **the model**. None is a relation between the counterfactual and **the process that generated the data**. A method can therefore score perfectly on all four while proposing a trajectory the world could never have produced, or an intervention that would in fact have left the outcome untouched. The evaluation cannot tell the two apart, because it never asks the world.

This is a measurement-validity problem rather than a leaderboard problem, and it is the reason the gap has persisted. Asking whether the world agrees requires a setting where the world is known: a benchmark that ships the mechanism that generated its own data, so that the *same* intervention a method claims credit for can be re-run through the true structural equations and the two outcomes compared. Real classification datasets cannot supply this. Synthetic temporal discovery benches ship a graph but no explanation method [Cheng et al., 2024, Herdeanu et al., 2025, Ferdous et al., 2025]. Temporal explanation benches put explainers in the loop but have no mechanism that could replay an intervention [Höllig et al., 2023b, Nguyen et al., 2024]. The closest evaluation of third-party counterfactuals against a known SCM is tabular [Höllig et al., 2023a].

We build that setting as a model–world benchmark. It generates multivariate time series from a known temporal structural causal model (SCM) with additive noise, which makes Pearl’s abduction step an exact subtraction [Pearl, 2009]: the exogenous noise of a given instance is recovered rather than approximated, so the counterfactual the world would have produced *for that instance* is computed exactly rather than estimated. (The invertibility of an additive-noise map is the ANM property of Hoyer et al., 2008; the abduction step itself is Pearl’s.) The benchmark scores a proposed counterfactual for mechanism consistency (CF-faith) and intervention size (do-complexity D), realises its implied intervention through the true mechanism, and decomposes the disagreement between what the explanation *claims* and what the world *does* into a trajectory term (the proposed trajectory is not the one the world would produce) and an outcome term (the classifier does not track the true label rule).

Designing the benchmark surfaced a construct-validity issue that we consider part of the contribution rather than an implementation detail. Its answer depends on how a proposal is *read*: interpreting a counterfactual as a single `do()` at one timestep inflates the trajectory term for densely-editing methods, while interpreting every edited slice as its own intervention drives that term to zero *by construction*. Neither endpoint is privileged, so we report **do-complexity** D , the number of timesteps a proposal must declare as interventions before the mechanism can reproduce it, alongside every disagreement number. A proposal with D close to the trajectory length is a trajectory rewrite under this benchmark; agreement obtained after declaring nearly every slice an action leaves little for the mechanism to predict.

A second finding is a reporting rule rather than a method comparison. Stationarity of a linear VAR requires a spectral radius below one [Lütkepohl, 2005], so an intervention’s effect on the label attenuates geometrically in the distance from the intervention time to the *label site*, not to the end of the trajectory. Those two distances coincide only when the label is terminal, as it is in LinearSCM-T, NlinearSCM-T, and SpringSCM-T, and in the spring experiments of Bahri et al. [2025], but not in UCR/UEA whole-series classification, where the discriminative site can be interior. Recourse feasibility is therefore a property of the benchmark configuration as much as of the method, and a validity number reported without its intervention-to-label distance is not interpretable. This is not a claim that recourse is impossible; it is a condition that every temporal recourse evaluation, including favourable ones, needs to state.

Contributions.

1. **A model–world benchmark** for temporal counterfactual explanations: the proposal’s own intervention is realised by the true mechanism, and the disagreement is decomposed additively into trajectory and outcome terms, each computed exactly against a known mechanism rather than estimated. Its construct validity is carried by do-complexity D , published beside every disagreement figure.
2. **Intervention-to-outcome distance as a required reporting condition.** The relevant distance is $t_{\text{label}} - t_0$, not $T - t_0$. The two coincide in the terminal-label families, so this contribution is the reporting rule, not an interior-label ablation.
3. **A benchmark suite with exact structural counterfactuals:** three additive-noise temporal SCM families shipping the true graph, the true mechanism, and oracle abduction–action–prediction counterfactuals, with the label functional exposed explicitly rather than hidden in the construction.
4. **A controlled demonstration of the measurement gap:** conventional metrics frequently remain favourable while structural replay reveals low mechanism consistency, large intervention schedules, and claimed target-class rates above the rates realised by the SCM.

90 **Scope.** Findings are established on LinearSCM-T, NlinearSCM-T, and SpringSCM-T, whose
 91 ground-truth mechanisms permit validation against the processes the benchmark claims to measure.
 92 The empirical results are descriptive point estimates for the registered benchmark realizations; they
 93 establish that conventional and mechanism-relative assessments can diverge, not a stable ordering of
 94 methods or mechanism families. Real-signal data could test whether the findings *travel*, but without
 95 a ground-truth mechanism they cannot validate the metrics, and no real-signal result is used here.

96 2 Related Work

97 **Counterfactual explanation for time series.** Methods span instance or segment substitution,
 98 saliency-guided optimisation, evolutionary search, and learned latent models [Delaney et al., 2021,
 99 Ates et al., 2021, Höllig et al., 2022, Li et al., 2024, Paredes Cetina et al., 2025, Yan and Wang,
 100 2023]. Work on plausibility and temporal coherence improves the proposal relative to observed
 101 data [Kostrzewa et al., 2026, Zuin and Veloso, 2026, Voets et al., 2026], but standard real-data
 102 benchmarks still cannot replay it through a known mechanism. CounTS performs abduction–action–
 103 prediction in its learned latent model; our benchmark instead uses the known data-generating process
 104 to evaluate an independently produced explanation. This distinction is not semantic: proximity asks
 105 whether the edit is small, plausibility whether it resembles observed data, and validity whether a
 106 model changes its decision. None asks whether the proposed intervention propagates through the
 107 process to the target label. A causally impossible trajectory can therefore remain conventionally
 108 successful.

109 **Benchmarks for temporal explanation.** XTSC-Bench and AMEE standardise temporal saliency
 110 evaluation, while CARLA systematises model-relative counterfactual success and distance [Höllig
 111 et al., 2023b, Nguyen et al., 2024, Pawelczyk et al., 2021]. Robustness and unjustified-counterfactual
 112 critiques sharpen those criteria but do not add a world-relative replay [Ghorbani et al., 2019, Yeh
 113 et al., 2019, Laugel et al., 2019, Poyiadzi et al., 2020]. The closest evaluation of third-party expla-
 114 nations against known SCMs is static and relation-based rather than a Pearl trajectory [Höllig et al.,
 115 2023a]. ExplainTS makes the real-data limitation explicit: without known relevance or mechanism,
 116 faithfulness is necessarily proxy-based [Mozolewski et al., 2026]. Our benchmark complements
 117 rather than replaces such suites. Their measures remain necessary for model-relative quality; the
 118 replay adds a missing estimand.

119 **Causal temporal benchmarks.** CausalTime, CausalDynamics, TimeGraph, and TCS expose
 120 lagged graphs to score structure recovery, not post-hoc explanations [Cheng et al., 2024, Herdeanu
 121 et al., 2025, Ferdous et al., 2025, Gkorgkolis et al., 2025]. DoFlow supports causal queries as a
 122 generative forecaster rather than a benchmark for third-party explanations [Wu et al., 2026]. The
 123 distinction is unit-level: drawing fresh noise estimates an interventional distribution, whereas retain-
 124 ing the abducted factual noise answers the counterfactual query [Thumm and Chen, 2026, Pearl,
 125 2009]. This is why a known graph alone is insufficient. The benchmark also needs the transition
 126 functions, the factual innovations, and an exposed label rule so that the same proposed action can be
 127 evaluated by both classifier and world.

128 **Causality used to constrain, versus causality used to derive.** TSCausalCF constrains a freely
 129 edited trajectory with local structural residuals [Bahri et al., 2025]; CounTS and causal recourse
 130 derive counterfactuals from learned or assumed models [Yan and Wang, 2023, Karimi et al., 2021].
 131 Our benchmark occupies a different role: the true temporal mechanism is an evaluation oracle for
 132 proposals generated independently of it. The TSCausalCF foil tests whether access to that mech-
 133 anism as a residual penalty is equivalent to explicit abduction–action–prediction. A local residual
 134 can be small at each edited slice without representing the trajectory produced by one intervention.
 135 Conversely, a structurally derived counterfactual is faithful to the model that generated it but cannot
 136 validate that learned model. The known-SCM benchmark separates these two uses of causality.

137 **Causal discovery.** Causal discovery solves the inverse problem of recovering structure [Pamfil
 138 et al., 2020, Runge et al., 2019, 2023]. Here the generator’s graph is ground truth and discovery
 139 checks remain outside the manuscript results. With an estimated graph, recovery error would be-
 140 come part of the benchmark error rather than a hidden property of the explainer.

141 3 The Model-vs-World Benchmark

142 The benchmark replays a proposal’s implied intervention in the known process for the same factual
 143 unit. Its evaluation protocol reports mechanism consistency, intervention size, and claimed-versus-
 144 realised target-class efficacy under a named intervention reading; it does not turn classifier validity
 145 into causal efficacy.

146 3.1 SCM, oracle, and identifying assumptions

147 For $\mathbf{x} = (\mathbf{x}_0, \dots, \mathbf{x}_{T-1})$, $\mathbf{x}_t \in \mathbb{R}^k$, the lagged SCM is

$$X_t^i = F_i(\text{Pa}_G(X_t^i)) + \varepsilon_t^i, \quad \text{Pa}_G(X_t^i) = \{X_{t-\ell}^j : G_{ij\ell} = 1, 1 \leq \ell \leq L\}. \quad (1)$$

148 The world label is $y_g(\mathbf{x}) = \mathbb{1}\{g(\mathbf{x}) > \theta\}$, whose final dependency is t_{label} ; h denotes the evaluated
 149 classifier.

150 Additivity makes Pearl abduction exact [Pearl, 2009, Hoyer et al., 2008]:

$$\varepsilon_t^i = x_t^i - F_i(\text{Pa}_G(x_t^i)). \quad (2)$$

151 For schedule $I = \{(t, S_t, \mathbf{a}_t)\}$, the oracle keeps the factual prefix, replaces intervened equations,
 152 and recursively evaluates the rest with the same abducted noise:

$$x_t^{I,i} = \begin{cases} a_t^i, & i \in S_t, \\ F_i(\text{Pa}_G(x_t^{I,i})) + \varepsilon_t^i, & i \notin S_t. \end{cases} \quad (3)$$

153 Fresh noise would instead answer an interventional query. Exactness requires the recorded G, F to
 154 be the data-generating SCM, positive lags, no unmeasured confounding, and additive innovations
 155 unchanged by I ; Appendix B records the derivation and claim boundary. The shared innovations are
 156 essential: they condition the replay on the same factual unit rather than comparing the proposal with
 157 another random trajectory. The oracle is therefore exact only inside the benchmark, where these
 158 identifying assumptions hold by construction.

159 3.2 CF-faith and do-complexity D

160 Given proposal $\hat{\mathbf{x}}$ and first change t_0 , CF-faith rejects retroactive edits and compares the suffix with
 161 either a zero-innovation deterministic rollout or the unit-level Pearl recursion in Eq. (3). These
 162 semantics answer different questions and are never averaged. Hard CF-faith is the fraction within
 163 tolerance; $t_0 \geq T - 1$ is unscorable because no suffix remains. The noiseless score asks whether
 164 the proposal follows the deterministic skeleton after its first action; the Pearl score asks whether
 165 it follows the unit-level continuation with factual innovations retained. A noisy proposal need not
 166 satisfy both. Publishing the scorable denominator prevents a terminal or empty edit from earning
 167 vacuous faithfulness.

168 Under Pearl semantics, D counts timesteps that must be imposed to reproduce $\hat{\mathbf{x}}$:

$$D(\hat{\mathbf{x}}; \mathbf{x}, F) = |\{t : S_t \neq \emptyset\}| \quad (4)$$

169 It counts timesteps, not coordinates: $D = 0$ is no action, $D = 1$ is compatible with an atomic
 170 intervention, and $D \approx T$ is a trajectory rewrite. Reading only the first slice can omit intended later
 171 actions; reading the complete schedule drives Δ_{traj} to zero by construction. We therefore publish D
 172 beside every disagreement. These endpoint readings expose complementary failures. A large first-
 173 slice trajectory error may mean that later edits were omitted from the action; a perfect full-schedule
 174 replay may mean that the method declared the whole trajectory. Neither low disagreement nor low
 175 D alone establishes a useful counterfactual.

176 3.3 Model-vs-world disagreement and its decomposition

177 For target class y^* , flip candidates are $\mathcal{C}_{y^*} = \{n : h(\mathbf{x}^{(n)}) \neq y^*\}$. Let I_n be the intervention read
 178 from an independently produced proposal $\hat{\mathbf{x}}^{(n)}$, and let $S_{y^*} = \{n \in \mathcal{C}_{y^*} : I_n \neq \emptyset\}$ be the scorable

179 population. We define

$$\begin{aligned}
A &= \mathbb{E}_{n \in \mathcal{S}_{y^*}} \mathbb{1}\{h(\hat{\mathbf{x}}^{(n)}) = y^*\}, & (\text{what the explanation claims}) \\
B &= \mathbb{E}_{n \in \mathcal{S}_{y^*}} \mathbb{1}\{h(\mathbf{x}^{I_n, (n)}) = y^*\}, & (\text{the model on the world's trajectory}) \\
C &= \mathbb{E}_{n \in \mathcal{S}_{y^*}} \mathbb{1}\{y_g(\mathbf{x}^{I_n, (n)}) = y^*\}, & (\text{what the world does}). \quad (5)
\end{aligned}$$

180 No-action proposals abstain; we report their fraction and the remaining denominator. This target-
181 class population estimates the sufficiency direction; PN and combined PNS are out of scope.

182 The signed disagreement decomposes exactly as

$$\underbrace{A - C}_{\Delta_{\text{total}}} = \underbrace{A - B}_{\Delta_{\text{traj}}} + \underbrace{B - C}_{\Delta_{\text{outcome}}}. \quad (6)$$

183 Δ_{traj} locates proposal-versus-replay disagreement; Δ_{outcome} locates classifier-versus-label-rule dis-
184 agreement on the replay. Positive Δ_{total} is over-claimed efficacy. We report CF-faith, D , and
185 the signed decomposition jointly, never as a scalar leaderboard. The intermediate rate B is what
186 makes the diagnosis possible: $A - B$ changes only the trajectory presented to the fixed classifier,
187 whereas $B - C$ changes only the decision rule applied to the replayed trajectory. The decomposition
188 therefore distinguishes an implausible proposal from classifier–world misalignment without turning
189 either term into a standalone ranking criterion.

190 3.4 The intervention-to-label reporting rule

191 For a stable linear process, an atomic intervention’s effect is bounded by a geometric factor in
192 $t_{\text{label}} - t_0$; Appendix B gives the derivation. Every validity and replay result must therefore report
193 the label functional, t_{label} , empirical t_0 , and label distance; multi-action proposals also require D . A
194 classifier flip without this distance is not evidence that the intervention could change the world. The
195 distance happens to equal $T - 1 - t_0$ for the terminal labels reported here, but trajectory end does
196 not define it. Methods acting at different times pose different recourse problems even on the same
197 factual population.

198 4 Benchmark design, methods, and controls

199 4.1 Generators as challenge factors

200 We report three additive-noise families: a contractive VAR (LinearSCM-T), a bounded masked-
201 MLP transition (NlinearSCM-T), and an undamped second-order particle system (SpringSCM-T).
202 All use $k = 10$, $L = 1$, $T = 100$, $N = 10,000$, Laplace innovations, and a terminal threshold label;
203 graph/coupling density is 0.2, 0.2, and 0.3. The families expose different transition structures, but
204 the comparisons do not identify mechanism family or contraction as a cause of observed differences.
205 Generator cards are in Appendix D. LinearSCM-T, NlinearSCM-T, and SpringSCM-T provide the
206 true graph G , mechanism F , factual noise, and label function, enabling exact structural replay and
207 validation of CF-faith, D , and $A/B/C$. LinearSCM-T is contractive by construction. NlinearSCM-
208 T is bounded but its spectral cap does not itself prove branch-level contraction. SpringSCM-T is
209 undamped and near-conservative. This variation challenges the same benchmark under different
210 dynamics; it is not a controlled ablation of contraction.

211 4.2 Methods and controls

212 One LSTM per family is held fixed across methods and shift evaluation (test accuracy 0.991,
213 0.999, and 0.994). The baselines are Wachter [Wachter et al., 2018], COMTE [Ates et al., 2021],
214 CONFETTI [Paredes Cetina et al., 2025], and M-CELS [Li et al., 2024]; none receives G or F .
215 TSCausalCF [Bahri et al., 2025] receives both as a residual-penalty foil. PearlSCMRecourse and
216 NoiselessSCMRecourse are structural controls (Appendix E). The controls optimise one action and
217 let either the Pearl recursion or the noiseless skeleton generate the suffix. They test the scoring
218 semantics; they are not competing post-hoc explainer.

Table 1: Model-relative counterfactual quality. V : validity; L_1 : proximity; S : unchanged-entry fraction; OOD: Isolation Forest score. L_1 and OOD are comparable only within families. †: foil; ‡: control.

family	method	$V \uparrow$	$L_1 \downarrow$	$S \uparrow$	OOD†
Linear	COMTE	1.00	30.33	.900	.027
Linear	M-CELS	.50	.29	.990	.025
Linear	CONFETTI	.56	93.98	.723	.027
Linear	Wachter	.90	344.88	.050	.028
Linear	TSCausalCF†	1.00	267.92	.003	.099
Linear	NoiselessSCMRecourse‡	1.00	192.46	.253	.077
Linear	PearlSCMRecourse‡	.64	2444.35	.498	-.088
Nlinear	COMTE	1.00	231.44	.794	-.021
Nlinear	M-CELS	.76	8.92	.926	.023
Nlinear	CONFETTI	.88	637.36	.560	-.074
Nlinear	Wachter	1.00	1299.36	.083	-.007
Nlinear	TSCausalCF†	.00	183.84	.006	.046
Nlinear	NoiselessSCMRecourse‡	.49	735.44	.404	-.022
Nlinear	PearlSCMRecourse‡	.58	642.62	.537	-.033
Spring	COMTE	1.00	1729.58	.810	.087
Spring	M-CELS	.31	50.62	.863	.080
Spring	CONFETTI	.64	2200.67	.676	.073
Spring	Wachter	.63	9951.79	.004	.071
Spring	TSCausalCF†	.05	1304.40	.007	.096
Spring	NoiselessSCMRecourse‡	1.00	2830.45	.271	.067
Spring	PearlSCMRecourse‡	1.00	2225.77	.260	.072

4.3 Evaluation protocol and reporting scope

Each configuration uses seed 42. We select the first 100 held-out instances predicted outside the target class and evaluate all seven methods on the same factu-als. The shift preserves graph, mechanism, seed, and classifier while replacing Laplace with uniform innovations. Tables are descriptive point estimates: they demonstrate disagreement between measurement systems, not stable rankings across SCMs or classifier fits. We report denominators and abstentions; construction diagnostics remain outside the manuscript. Shift-VR is shifted/base validity when base validity is at least 0.30; below that threshold both rates are retained and the unstable ratio is suppressed. No smoke configuration or dataset-level diagnostic enters a paper table.

5 Benchmark Results

The same proposals receive a conventional model-relative reading and a structural replay; the contrast demonstrates a measurement gap rather than an inferential ranking.

5.1 Conventional metrics suggest successful counterfactual generation

Without the mechanism, Table 1 is reassuring: baseline validity is at least 0.50 in 11/12 cells and at least 0.80 in six. COMTE always flips the classifier; M-CELS makes the smallest and sparsest edits with lower validity; Wachter is denser. These are conventional trade-offs, not evidence that an intervention would work. The same reading would also favour different methods depending on which conventional criterion is prioritised. COMTE dominates validity but not proximity, M-CELS is the closest and sparsest baseline, and the OOD score does not provide a consistent ordering across families. No conventional column identifies whether the edited suffix follows the transition equations or whether the target label is preserved after structural replay.

Ten of 12 baseline Shift-VR values are at least 1; M-CELS and CONFETTI on NlinearSCM-T give 0.93 and 0.86. Thus the innovation shift generally leaves classifier-flip rates intact and sometimes improves them. This is a useful robustness statement about the explainer-classifier pair, but it does not test causal efficacy: a proposal can remain equally successful under the shift while disagreeing with both base and shifted mechanisms in the same way.

Controls anchor the intervention semantics. NoiselessSCMRecourse has rollout/Pearl CF-faith 1/0 in every family; PearlSCMRecourse has 0/1, $D = 1$, and $\Delta_{\text{total}} = 0.29, 0.00, 0.02$. Re-

Table 2: Innovation-shift robustness. Cells give base/shifted validity/Shift-VR; the ratio is suppressed when base validity is below 0.30.

method	Linear	Nlinear	Spring
COMTE	1.00/1.00/1.00	1.00/1.00/1.00	1.00/1.00/1.00
M-CELS	.50/.63/1.26	.76/.71/.93	.31/.31/1.00
CONFETTI	.56/.60/1.07	.88/.76/.86	.64/.86/1.34
Wachter	.90/.93/1.03	1.00/1.00/1.00	.63/.70/1.11
TSCausalCF [†]	1.00/1.00/1.00	.00/.00/–	.05/.09/–
NoiselessSCMRecourse [‡]	1.00/1.00/1.00	.49/.40/.82	1.00/1.00/1.00
PearlSCMRecourse [‡]	.64/.75/1.17	.58/.45/.78	1.00/1.00/1.00

Table 3: Mechanism-relative benchmark results. R/P: rollout/Pearl hard CF-faith; n_s : scorable count; D : Pearl do-complexity; $A/B/C$: proposed-model / replayed-model / replayed-world rates. Last: $\Delta_{\text{traj}}/\Delta_{\text{outcome}}/\Delta_{\text{total}}$. *: threshold-sensitive; †: foil; ‡: control.

family	method	R/P-faith	n_s	D	$A/B/C$	$\Delta_{\text{traj}}/\Delta_{\text{outcome}}/\Delta_{\text{total}}$
Linear	COMTE	.00/.00	100	100.0	1.00/.00/.00	1.00/.00/1.00
Linear	M-CELS	.00/.075	53	6.7*	.94/.00/.00	.94/.00/.94
Linear	CONFETTI	.00/.00	56	100.0	1.00/.00/.00	1.00/.00/1.00
Linear	Wachter	.00/.00	100	98.3	.90/.00/.00	.90/.00/.90
Linear	TSCausalCF [†]	.55/.00	100	100.0	1.00/.00/.00	1.00/.00/1.00
Linear	NoiselessSCMRecourse [‡]	1.00/.00	100	75.0	1.00/.00/.00	1.00/.00/1.00
Linear	PearlSCMRecourse [‡]	.00/1.00	100	1.0	.64/.64/.35	.00/.29/.29
Nlinear	COMTE	.00/.00	100	100.0	1.00/.16/.16	.84/.00/.84
Nlinear	M-CELS	.00/.24	100	26.4*	.76/.00/.00	.76/.00/.76
Nlinear	CONFETTI	.00/.00	88	100.0	1.00/.19/.19	.81/.00/.81
Nlinear	Wachter	.00/.00	100	100.0	1.00/.43/.43	.57/.00/.57
Nlinear	TSCausalCF [†]	.00/.00	100	100.0	.00/.36/.36	–.36/.00/–.36
Nlinear	NoiselessSCMRecourse [‡]	1.00/.00	100	59.7	.49/.35/.35	.14/.00/.14
Nlinear	PearlSCMRecourse [‡]	.00/1.00	87	1.0	.67/.67/.67	.00/.00/.00
Spring	COMTE	.00/.00	100	100.0	1.00/.03/.03	.97/.00/.97
Spring	M-CELS	.00/.40	100	55.3*	.31/.00/.01	.31/–.01/.30
Spring	CONFETTI	.00/.00	64	100.0	1.00/.313/.297	.688/.016/.703
Spring	Wachter	.00/.00	100	100.0	.63/.38/.38	.25/.00/.25
Spring	TSCausalCF [†]	.00/.00	100	100.0	.05/.04/.04	.01/.00/.01
Spring	NoiselessSCMRecourse [‡]	1.00/.00	100	73.0	1.00/.49/.48	.51/.01/.52
Spring	PearlSCMRecourse [‡]	.00/1.00	100	1.0	1.00/1.00/.98	.00/.02/.02

247 producing the noiseless continuation under Pearl semantics instead requires $D = 75.0, 59.7, 73.0$.
 248 The controls recover their construction and show why the intervention reading must be named. In
 249 particular, the noiseless control is not a failed causal method when its Pearl score is zero: it intention-
 250 ally answers a different counterfactual question. Conversely, the Pearl control’s unit do-complexity
 251 and perfect Pearl faithfulness show that the benchmark does not penalise every long changed suffix
 252 merely because many values differ from the factual. It distinguishes values generated by propagation
 253 from values that must be imposed.

254 **Mechanism consistency is largely absent.** For the four baselines, hard rollout CF-faith is zero
 255 in all 12 cells and hard Pearl CF-faith is zero in nine. Only M-CELS has nonzero Pearl fractions
 256 (0.075, 0.24, 0.40), so favourable validity and sparsity do not imply an SCM-consistent suffix. This
 257 is the first reversal of the conventional account: the methods often find inputs the classifier accepts,
 258 but almost none of those inputs is the continuation generated by one declared action and the factual
 259 innovations. CF-faith is an admission check rather than a ranking: zero marks a mismatch with the
 260 named semantics, while a nonzero fraction still requires D and outcome replay to determine what
 261 was admitted.

262 **Replay locates the disagreement.** $A > C$ in every baseline cell ($\Delta_{\text{total}} = 0.25\text{--}1.00$). Moreover,
 263 $\Delta_{\text{outcome}} = 0$ in 10/12 cells and is only $-0.01, 0.02$ in the others; Δ_{traj} therefore accounts for
 264 nearly all disagreement. The observed gap is between proposal and structural replay, not classifier
 265 and label rule on the replay. This localization matters. It means that the headline failure cannot
 266 generally be repaired by recalibrating the fixed LSTM on trajectories that the world actually pro-
 267 duces: on those replays, the classifier and the known label rule nearly always agree. Instead, the
 268 classifier is evaluated on a proposed trajectory that the corresponding intervention does not realise.

269 The benchmark therefore changes the empirical question from “did the edited input flip the model?”
 270 to “did the stated action generate the edited input, and did its propagated consequences retain the
 271 target outcome?”

272 **Do-complexity reveals trajectory rewrites.** COMTE and scorable CONFETTI proposals have
 273 $D = 100$; Wachter has 98.3, 100, 100. They are near-complete schedules under Pearl semantics.
 274 M-CELS has $D = 6.7, 26.4, 55.3$, but these values change $6.1\text{--}16.5\times$ under the registered thresh-
 275 old perturbation. Only the ordinal “less complete” comparison is supported, and its claimed rate
 276 still exceeds replay in every family. This prevents a misleading interpretation of low replay error
 277 under a full schedule: if every timestep is declared an action, reproducing the proposal says little
 278 about causal propagation. COMTE and CONFETTI illustrate this vacuity directly. M-CELS inter-
 279 venes less extensively, but the threshold sensitivity prevents treating its exact D values as stable
 280 measurements.

281 **Residual regularisation is not structural derivation.** TSCausalCF has $D = 100$ and zero Pearl
 282 CF-faith in all families. On LinearSCM-T it has validity 1, rollout faith 0.55, but replay rate 0; on
 283 NlinearSCM-T and SpringSCM-T, $\Delta_{\text{total}} = -0.36, 0.01$. Thus mechanism access through a local
 284 residual is not equivalent to explicit structural derivation. The nonlinear negative disagreement also
 285 cautions against interpreting the smallest absolute Δ_{total} as the best explanation: its proposed target-
 286 class rate is zero, and the method is not solving the same successful-recourse problem as COMTE.
 287 The components and the conventional validity must remain visible together.

288 **Timing and populations qualify comparisons.** All labels are at $t_{\text{label}} = 99$. COMTE, CON-
 289 FETTI, and TSCausalCF start nonempty proposals at $t_0 = 0$; Wachter is at 0 except for a few Linear
 290 cases up to 18. M-CELS median label distances are 6, 67.5, 95; control medians are 74, 49, 74 (noise-
 291 less) and 49, 49, 74 (Pearl). These differences qualify rather than explain performance. Population
 292 also matters: Linear M-CELS has validity .50 but $A = .94$ on 53 scorable proposals; CONFETTI
 293 has 56, 88, 64 scorable, and Nlinear PearlSCMRecourse has 87.

294 **Joint interpretation.** Read jointly, conventional validity and shift retention establish that the base-
 295 lines often achieve their model-relative objectives, while the controls show that the benchmark recog-
 296 nises structural continuations under matching semantics. Baseline proposals nevertheless rarely pass
 297 mechanism consistency, and their target-class rates fall after replay. The decomposition assigns al-
 298 most all of that fall to the proposed trajectory, while D shows that a low full-schedule error would
 299 often require declaring nearly every timestep an intervention. CF-faith, D , and $A/B/C$ are therefore
 300 complementary: they test what follows the mechanism, what must be imposed, and what outcome
 301 the replay preserves. Conventional metrics remain useful, but they do not identify the causal object
 302 implied when an edited trajectory is presented as recourse.

303 6 Conclusion

304 Conventional validity is a statement about a classifier, not evidence that an intervention would pro-
 305 duce the proposed trajectory or outcome. A known additive-noise temporal SCM makes that dis-
 306 tinction measurable: exact replay separates mechanism consistency, intervention size, trajectory
 307 disagreement, and classifier–world disagreement. In the three reported benchmark realizations, con-
 308 ventional metrics frequently remain favourable while the standard baselines have little hard CF-faith,
 309 often require large intervention schedules, and claim target-class rates above those obtained by struc-
 310 tural replay. The $A/B/C$ decomposition locates nearly all of this observed gap at the trajectory
 311 level.

312 These synthetic results establish a measurement failure that model-relative metrics cannot detect;
 313 they do not establish method rankings or real-world causal validity. A temporal counterfactual
 314 evaluation should therefore report conventional metrics together with both CF-faith semantics, do-
 315 complexity D , $(\Delta_{\text{traj}}, \Delta_{\text{outcome}}, \Delta_{\text{total}})$, the benchmark population, and $t_{\text{label}} - t_0$. Progress under
 316 this contract means proposals that are mechanism-consistent, require restrained intervention sched-
 317 ules, and have small $|\Delta_{\text{total}}|$ at a stated and feasible label distance.

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411 A Limitations

412 Exactness is conditional on the written G, F being the additive-noise data-generating SCM, positive
 413 lags, recoverable factual noise, and no unmeasured confounding; the presets also fix $L = 1$. A
 414 discovered, misspecified, or non-additive mechanism changes the estimand. The benchmark also de-
 415 pends on the intervention reading and changed-coordinate tolerance, so CF-faith, D , and Δ must be
 416 interpreted jointly. M-CELS’s threshold sensitivity supports only an ordinal schedule comparison.

417 All labels are terminal, so these experiments do not test an interior discriminative site; differing em-
 418 pirical t_0 also prevents attributing method gaps to label distance alone. Finally, the point estimates
 419 do not quantify variation across independently generated SCMs or classifier fits. They demonstrate
 420 that measurement systems can disagree, not stable rankings, family-level causes, or real-world va-
 421 lidity. We propose an evaluation benchmark, not a counterfactual generator.

422 The intervention schedule is reconstructed from changed coordinates using a 10^{-3} tolerance. That
 423 rule is explicit and shared across metrics, but it is not uniquely mandated by causal semantics. A
 424 method could regard a dense optimised trajectory as one high-level action, while the benchmark can
 425 only evaluate actions expressible in its observed variables and structural equations. Our endpoint
 426 readings therefore diagnose the cost of two transparent interpretations; they do not settle which
 427 action language is appropriate for a deployment domain. The strong threshold sensitivity for M-
 428 CELS shows why this choice must be reported rather than hidden.

429 Finally, all three families are synthetic, additive-noise, fully observed, and use the same terminal
 430 threshold label. This enables exact unit-level replay but excludes latent confounding, measurement
 431 error, learned mechanisms, non-invertible noise, and labels that depend on interior or distributed
 432 trajectory features. The experiment also uses one generated benchmark and one classifier fit per
 433 family. Additional seeds would be required for uncertainty intervals and method comparisons; real
 434 data with an estimated SCM would conflate explanation error with mechanism-estimation error.
 435 These limitations narrow the positive claim to the benchmark’s construct validity and the observed
 436 conventional-causal disagreement in the reported realizations.

437 B Benchmark derivation and claim boundary

438 The oracle in Eq. (3) is a unit-level counterfactual because it reuses the innovations abducted from
 439 the factual trajectory. Drawing new innovations would instead estimate $P(\mathbf{X} \mid \text{do}(I))$. Exactness
 440 requires the recorded graph and mechanism to be the data-generating SCM, positive-lag causes,
 441 no unmeasured confounding, additive innovations, and an intervention that does not alter those
 442 innovations. Relaxing additivity requires an identified inverse noise model; replacing the written
 443 mechanism with a discovered one makes discovery error part of the estimand.

444 For an atomic intervention at t_0 , let $\delta_t = \mathbf{x}_t^I - \mathbf{x}_t$. In the linear $L = 1$ family, $\delta_t = A\delta_{t-1}$ after the
 445 action. Stationarity requires $\rho(A) < 1$ [Lütkepohl, 2005]; Gelfand’s formula implies that for any
 446 $r \in (\rho(A), 1)$ there is finite K with

$$\|\delta_{t_{\text{label}}}\| \leq Kr^{t_{\text{label}}-t_0}\|\delta_{t_0}\|. \quad (7)$$

447 Contractive nonlinear transitions give the analogous product of local Lipschitz factors. Thus the
 448 relevant distance is to the label site. If $t_0 > t_{\text{label}}$, the action cannot affect the label; for terminal
 449 labels, $T - 1 - t_0$ merely happens to equal that distance.

450 C Metric definitions and reporting conventions

451 The validity argument is in Section 3. This appendix records populations and the numerical con-
 452 ventions needed to reproduce the reported model-relative, innovation-shift, and mechanism-relative
 453 quantities. CF-faith, D , and the disagreement terms are interpreted jointly, never averaged into a
 454 composite score. A hard CF-faith score of 0 does not remove a method from the reported popula-
 455 tion.

456 **Changed-element predicate.** A coordinate is changed when $|\hat{x}_t^i - x_t^i| > 10^{-3}$. That single thresh-
 457 old defines t_0 (the first changed timestep), CF-faith’s retroactive gate, sparsity, and the action set S_t
 458 inside D . If no coordinate exceeds it, $t_0 = T - 1$ and the instance is CF-faith-degenerate.

Model-relative counterfactual quality. Validity is the fraction of proposals with $h(\hat{\mathbf{x}}) = y^*$. It is reported with t_{label} , t_0 , and $t_{\text{label}} - t_0$; without that distance it is not interpretable (Section 3). Proximity is $\|\hat{\mathbf{x}} - \mathbf{x}\|_p$ over the flattened $T \times k$ trajectory, $p \in \{1, 2\}$. Sparsity is the fraction of the $T \times k$ entries left *unchanged* ($|\Delta| \leq 10^{-3}$). Higher means a sparser edit; 1 means the proposal equals the factual. This is the complement of an L_0 altered-fraction; we emit that complement as `frac_altered` and do not treat the two as interchangeable. Channel and timepoint variants are the fractions of channels (resp. timesteps) left entirely untouched. OOD plausibility is the Isolation Forest decision score on the flattened trajectory (higher = more in-distribution). TRSI is implemented as a mechanism-free smoothness descriptor and is not reported here.

CF-faith, both semantics. Given $\hat{\mathbf{x}}$ and t_0 , CF-faith fails if any prefix coordinate is changed (retroactive edit). Otherwise it compares the suffix with one of two continuations of F , never averaged. Under `noiseless_rollout` the reference is the deterministic skeleton (subsequent innovations set to zero). Under `pearl_delta` it is the abduction–action–prediction recursion of Eq. (3). A proposal in a noisy SCM is not expected to be both. The residual is the mean L_1 distance between the proposed suffix and the reference; the hard score is 1 if that residual is below 10^{-3} and 0 otherwise. The soft score $\exp(-\text{residual})$ is the graded companion. This paper reports the hard fraction. When $t_0 \geq T - 1$ both scores are NaN, not 1: there is no suffix to test. Aggregates use the scorable subset and are published with `frac_degenerate` and `n_cf_faith_scorable`.

Do-complexity and populations. D is Eq. (4), counted on timesteps, under Pearl semantics unless a noiseless reading is named. $D = 0$ is a no-action proposal; `frac_vacuous` is that row of the same distribution. Flip candidates are $\mathcal{C}_{y^*} = \{n : h(\mathbf{x}^{(n)}) \neq y^*\}$. The scorable population $\mathcal{S}_{y^*}$ further drops empty schedules. No-action proposals abstain from Eq. (5); we report their fraction and the remaining denominator rather than scoring them as correct zero-effect interventions. Because $\mathcal{S}_{y^*}$ conditions on the model being outside the target class, the estimand is the sufficiency direction. PN, PS, and their PNS combination are defined in the protocol and are outside this paper’s scope.

Sign of Δ . $\Delta_{\text{total}} = A - C$ is signed. Positive means over-claimed efficacy (the explanation asserts a target-class outcome the world denies); negative means the classifier misses an effect present under the label rule y_g . Distance from zero measures disagreement; the sign is retained for diagnosis. Δ_{traj} and Δ_{outcome} locate the source and are not a second leaderboard. A Δ of either kind is never cited without D and the intervention reading.

Innovation-shift robustness. Shift-VR is validity retention on a held-out environment of the same SCM (frozen classifier): $\text{validity}_{\text{shift}}/\text{validity}_{\text{base}}$ when base validity is at least 0.30. It is reported in Table 2; below that threshold, the base and shifted rates are shown but the unstable ratio is suppressed. Input sensitivity is a Lipschitz constant of the explanation in the input, where exposed, and is not reported.

D Generator specifications

The paper reports three families: LinearSCM-T, NlinearSCM-T, and SpringSCM-T. Other generator scales exist in the code for plumbing and tests and are not reported.

All three families share the same outer protocol. A lagged graph $G \in \{0, 1\}^{k \times k \times L}$ is drawn with independent Bernoulli(s) edges, except that lag-1 self-loops are forbidden. Innovations are additive Laplace with scale 0.1. A burn-in of 100 steps is discarded. The binary label is a median split, over the generated population, of channel 0 at $t_{\text{label}} = T - 1$, and the resulting threshold is recorded with the benchmark artifacts. Observation is the identity: nonlinear mixing $x = g(z)$ is out of scope, because it would make abduction partial.

LinearSCM-T. A stable VAR,

$$x_t = \sum_{\ell=1}^L A_{\ell} x_{t-\ell} + \varepsilon_t. \quad (8)$$

Entries of each A_{ℓ} are drawn from $U(-0.5, 0.5)$, masked by G , and rescaled so that $\rho(A_{\ell}) \leq 0.9$. With $L = 1$ this is the contraction that Eq. (7) uses.

506 **NlinearSCM-T.** The same G and additive noise, with the linear map replaced by a per-node two-
507 layer MLP

$$x_t^i = \text{decay}_i x_{t-1}^i + \text{gain} \cdot \tanh(\text{MLP}_i(\text{masked lags})) + \varepsilon_t^i. \quad (9)$$

508 Registered hyperparameters are hidden width 16, gain = 1.5, decay_{*i*} $\sim U[0.1, 0.4]$, spectral-norm
509 cap 6.0 on each weight matrix, and init-gain 3.0. Masked inputs are scaled by $1/\sqrt{\max(n_{\text{parents}}, 1)}$.
510 Boundedness follows from gain $\cdot \tanh$ and decay < 1 ; the cap of 6.0 does not itself make the
511 nonlinear branch Lipschitz-contractive, so dissipation is an empirical property of the family rather
512 than a constructive one. A randomly initialised two-layer net stays close to its own linearisation,
513 so this family is a nonlinear *parameterisation* of an additive-noise transition, not a strongly curved
514 mechanism.

515 **SpringSCM-T.** An undamped second-order particle system, adopted as used by Bahri et al. [2025].
516 Five particles are exposed as $k = 10$ channels (position then velocity). One symplectic-Euler step
517 with spring constant 0.3, $\Delta t = 0.1$, and unit mass gives

$$\begin{aligned} a_t^i &= - \sum_j K_{ij} (x_t^i - x_t^j), \\ v_{t+1}^i &= v_t^i + \Delta t a_t^i, \\ x_{t+1}^i &= x_t^i + \Delta t v_t^i. \end{aligned} \quad (10)$$

518 Additive innovations are then applied as in the other families. K is 0.3 times a particle-level
519 Bernoulli(0.3) mask with no self-coupling; the last two particles have their incoming edges zeroed,
520 so two exogenous particles exist by construction. There is no damping term. The recorded graph
521 stores only the velocity-from-position couplings; the position-from-velocity self-coupling is treated
522 as baseline structure and omitted. The label is the same channel-0 median threshold as the other
523 families (first-particle position), not an energy functional.

Table 4: Paper generator families. Every row uses $L = 1$, a terminal label, and Laplace innovations. For SpringSCM-T, s is a particle-level coupling probability.

family	mechanism	k	T	N	s
LinearSCM-T	VAR	10	100	10,000	0.2
NlinearSCM-T	MLP	10	100	10,000	0.2
SpringSCM-T	spring	10	100	10,000	0.3

524 **Registered deviations.** The original design space listed $k \in \{5, 10, 20\}$, $L \in \{1, 3\}$, $s \in$
525 $\{0.1, 0.2, 0.3\}$, $T \in \{50, 100, 200\}$, and a segment count. The three families lock $k = 10$, $L = 1$,
526 $T = 100$, $N = 10,000$. Changing L would invalidate the per-matrix spectral rescaling and is
527 not a casual configuration edit. Segments are unused. KuramotoSCM-T was removed. Multi-seed
528 replicates of a family change only the seed.

529 E Control methods

530 PearlSCMRecourse and NoiselessSCMRecourse are positive controls constructed by this work, not
531 adapters of a published explainer. Both receive the true mechanism F . Both leave the factual prefix
532 unchanged, optimise a single slice $\mathbf{x}_{t_0} + \delta$, and let F produce the suffix. A non-vacuous proposal
533 therefore has $D = 1$ by construction. Candidate intervention times are $t_0/T \in \{0.25, 0.5\}$. The
534 search minimises

$$\lambda_{\text{pred}} \text{CE}(h(\hat{\mathbf{x}}), y^*) + \lambda_{\text{prox}} \|\delta\|_2^2 \quad (11)$$

535 with Adam over $\delta \in \mathbb{R}^k$ (300 steps for NoiselessSCMRecourse, 500 for PearlSCMRecourse). If that
536 search does not produce a genuine above-tolerance flip, one prediction-only retry with $\lambda_{\text{prox}} = 0$ is
537 allowed. Registered weights are $\lambda_{\text{pred}} = 1$ for both controls, $\lambda_{\text{prox}} = 0.1$ for PearlSCMRecourse
538 and $\lambda_{\text{prox}} = 0.5$ for NoiselessSCMRecourse.

539 **PearlSCMRecourse.** The suffix is the abduction–action–prediction recursion of Eq. (3):

$$\hat{x}_t = \begin{cases} x_t, & t < t_0, \\ x_{t_0} + \delta, & t = t_0, \\ F(\text{Pa}_G(\hat{\mathbf{x}}_t)) + \varepsilon_t, & t > t_0, \end{cases} \quad (12)$$

540 with ε_t abducted from the factual instance as in Eq. (2). Hard CF-faith under `pearl_delta` is
 541 therefore 1 by construction, provided the instance is scorable ($t_0 < T - 1$).

542 **NoiselessSCMRecourse.** The same search, with all subsequent innovations set to zero:

$$\hat{x}_t = \begin{cases} x_t, & t < t_0, \\ x_{t_0} + \delta, & t = t_0, \\ F(\text{Pa}_G(\hat{\mathbf{x}}_t)), & t > t_0. \end{cases} \quad (13)$$

543 Hard CF-faith under `noiseless_rollout` is 1 by construction on scorable instances. The two
 544 controls are not interchangeable: a noiseless skeleton is not the unit-level Pearl counterfactual, and
 545 we never average the two CF-faith semantics.